

Testosterone Replacement Therapy (Men)

Patient Information · Treatment for confirmed low testosterone with symptoms

WARWIKI

Testosterone replacement therapy (TRT) is for men who have **both** a confirmed low testosterone level **and** symptoms, such as low sex drive or low energy. It brings testosterone back to a normal range. It is **not** an anti-aging treatment, and it is not for men with normal levels.

Who It Is For

Low testosterone is found with **two morning blood tests** plus symptoms. One low number alone is not enough.

What it can help: sex drive and sexual activity improve most. Energy and mood may improve a little. It can also help muscle, bone density and some types of anemia. **On its own, it does little for erection problems**, but when levels are low it may help erection pills work better.

The Forms

- **Gel or liquid** on the skin every day. Others must not touch the skin where you put it.
- **Shots** in the muscle or under the skin, weekly or every 2 weeks. A long-acting shot is given every 10 weeks after the first two; you stay in the office for 30 minutes after each one.
- **Capsules** taken twice a day with food.
- **Pellets** placed under the skin every 3–6 months, or a **nasal gel** used 3 times a day.

Heart and Other Risks

A large study found that testosterone gel did not raise the risk of heart attack or stroke in men at higher heart risk. It did show slightly more blood clots in the lungs, an irregular heartbeat (atrial fibrillation) and broken bones. Another study found the risk of blood clots was highest in the first 6 months.

Learn the Terms

Testosterone

The main male hormone. It affects sex drive, energy, muscle and bone.

Hypogonadism

A confirmed low testosterone level.

Hematocrit

A blood test that shows how thick the blood is. TRT can raise it.

PSA

A blood test for the prostate, checked before and during treatment.

Fertility

The ability to father a child. TRT lowers sperm production.

Gel transfer

Gel rubbing off onto another person's skin.

Planning to have children? Tell your team first. TRT **lowers sperm production**. It should not be used if you want children now or in the future. After stopping, sperm may take months or longer to come back, and in some men it does not fully recover. If you are not sure, ask about freezing sperm before you start. Other treatments can raise testosterone without this effect.

Check-Ups and Blood Tests

You will have a visit at about 3–6 months, then every 6–12 months. Your team checks:

- **Testosterone level**, to find the right dose
- **Hematocrit**. If the blood gets too thick, the dose is lowered or paused.
- **PSA** for the prostate, in men over 40
- **Blood pressure**, before and during treatment

Tell Your Team If You Have Had

- A heart attack or stroke in the past 6 months, or heart failure
- Blood clots, or a family history of blood clots
- Sleep apnea that is not treated
- Prostate or breast cancer, or trouble urinating

Using It Safely

- **Gel**: put it only where the label says, let it dry, cover the area and wash your hands. Keep women and children away from the skin where you put it.
- Do not change your own dose. Keep all lab visits.
- If you take warfarin or insulin, your team may need to check or change it when you start or stop TRT.

Call the office that is treating you if you have:

- New trouble urinating
- Snoring or sleep problems that get worse
- High blood pressure readings at home
- New or worse low mood, anxiety or other mood changes

Go to an emergency department if:

- You have chest pain, trouble breathing, or a racing or uneven heartbeat
- One leg is swollen, warm or painful
- You cough, or your throat feels tight, after a long-acting shot
- You have thoughts of harming yourself
- You cannot reach the office and are worried

Three Things to Remember

1. TRT is for men with confirmed low testosterone and symptoms.
2. It lowers sperm production. Tell your team first if you want children now or later.
3. Regular blood tests and blood pressure checks are part of treatment.